

The Shortest Superpermutation on 6 Symbols has Length 872

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Abstract

A superpermutation on n symbols is a string containing every permutation of the symbols as a contiguous window, and $M(n)$ is the least length of one. We prove that no superpermutation on six symbols has length 871 or less; with the known string of length 872 this gives an independent proof that $M(6) = 872$. An exact accounting identity turns length 871 into a structural budget. Counting constraints valid for every superpermutation force a candidate of that length to have zero slack and place it in one of sixteen structural classes. Exhaustive, independently cross-checked searches then show that every class is empty. The result is computer assisted: every computational step is carried by a named program in an accompanying artifact repository, and the whole computational core re-runs on one laptop inside a working day.

1 Introduction

A *superpermutation* on the alphabet $\{1, \dots, n\}$ is a string over that alphabet containing each of the $n!$ permutations as a window of n consecutive characters, and $M(n)$ is the least length of such a string. The string 121 shows $M(2) \leq 3$ and 123121321 shows $M(3) \leq 9$, and both are optimal. The problem was posed by Ashlock and Tillotson [1], whose recursive construction gives $M(n) \leq 1! + 2! + \dots + n!$ and who conjectured that bound exact: it holds for $n \leq 5$, where $M(1), \dots, M(5) = 1, 3, 9, 33, 153$, the last value settled by the exhaustive search of Chaffin [2], and it fails at $n = 6$, where Houston [3] produced in 2014 a superpermutation of length 872 against the conjectured 873, so $M(6) \leq 872$. The survey of Engen and Vatter [4] is the peer-reviewed account of the classical background and of the earlier lower bounds.

Whether 872 was optimal remained open for more than a decade. The general lower bound of the anonymous 4chan poster, Houston, Pantone and Vatter [5], $M(n) \geq n! + (n-1)! + (n-2)! + n - 3$, which is Theorem 2.5 of [4], gives $M(6) \geq 867$; Houston's improvement by one, recorded in the remark following that theorem, gives $M(6) \geq 868$. In 2026 Raudvere [6] gave a machine-checked Lean 4 proof of $M(6) \geq 868$, the first publicly available machine-checked proof of Houston's +1, and Hunter and Raudvere [7] a machine-checked general bound giving $M(6) \geq 869$. The residual gap consists of the lengths 869, 870 and 871, and is the gap addressed here.

Fritsch [8] announced a proof of minimality on the Superpermutators mailing list in 2021, pairing a lower-bound argument with a search at length 871; the public record describes that proof as unfinished and, to our knowledge, no completed version has appeared, so we do not treat the announcement as a completed proof. The strategy followed here, a structural lower bound sharpened until an exhaustive elimination at length 871 becomes affordable, is the same one. In 2026 the question drew a computer-assisted proof package [9], developed by a different group on a different decomposition, and a mailing-list thread [10] announcing a Lean-verified candidate. The present work was developed independently of both.

The work was carried out in three phases. In the first, the model that produced the proofs and the programs had no network access and worked from the definitions alone; the searches, the classification and the closure-topology theorem were complete and had passed an internal adversarial review chain by 4 September

*The research reported here was directed by Worasait Suwannik. The proofs and the programs were produced by Claude, Anthropic's AI model.

2026. In the second, the author relayed three rounds of external referee reports; in response the proofs of the scanner inequalities, the cycle-shape exclusion and the coverage theorem were written out in the form given here. In the third, from 7 September, the author supplied the package [9] with its paper. Its literature review is the source of the priority chain and of the citations [6, 7]; its published length-872 witness is the external control of Section 5.3; and its list of open obligations prompted the clean-room second implementation of the loop-mode search and the second derivation of the one certified capacity value. The thread [10] was shown on 8 September as an index entry. No mathematical statement in this paper derives from either.

The argument reduces the existence of a length-871 superpermutation to the acceptance of finitely many exhaustive searches, and the reduction itself is carried by lemmas proved from the definitions, all of which are given here. A structural class is called *empty* when no first-occurrence ordering whatsoever lies in it, not merely when some search failed to find one; every emptiness claim in this paper is of that universally quantified kind, and that is what makes an exhausted search a proof. The programs, run logs and certificates are in an artifact repository [11], and Section 5 documents each program in turn.

2 From strings to a single finite question

Theorem 2.1 (Main Theorem). *No superpermutation on $\{1, \dots, 6\}$ has length at most 871; that is, $M(6) \geq 872$.*

Proposition 2.2 (Upper bound). $M(6) \leq 872$.

Proof. Houston [3] exhibits a superpermutation on six symbols of length 872. Such a string is a certificate, checked by sliding a width-6 window along it and counting the distinct windows that are permutations; one is shipped as `data/best872.txt` in [11] and checked in under a minute by the program of Section 5.1. $\square$

Corollary 2.3. $M(6) = 872$.

The proof of Theorem 2.1 occupies Sections 2 to 4, and Figure 1 is its map. This section turns the theorem into one finite question about one length.

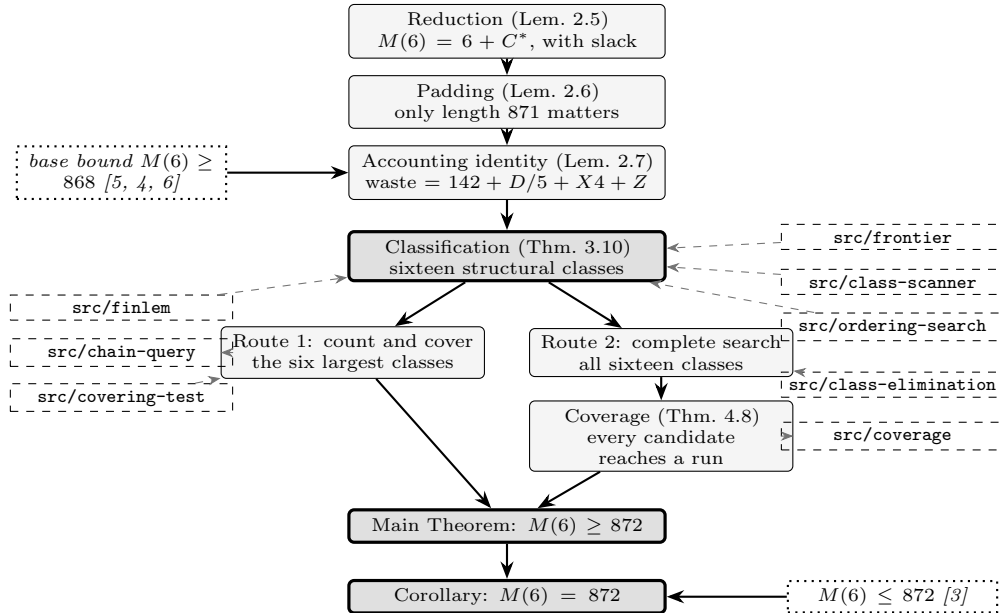


Figure 1: How the pieces connect. Shaded boxes are the statements this paper proves, the two heavy ones being the principal results; dotted boxes are taken from the literature; dashed boxes are the programs of Section 5, attached by dashed arrows to the statement whose exhaustive enumeration they carry. The upper bound enters only at the corollary, so the whole spine below the reduction is the lower bound.

2.1 The reduction

Definition 2.4. For distinct permutations $p \neq q$ of $\{1, \dots, 6\}$, written as six-letter words, let $d(p, q)$ be 6 minus the length of the longest suffix of p that is also a prefix of q . Since $p \neq q$ that suffix is proper, so $d(p, q) \in \{1, \dots, 6\}$, and $d(p, q)$ is the number of characters that must be appended after an occurrence of p to produce an occurrence of q . The overlap graph has the 720 permutations as vertices and an arc $p \rightarrow q$ of weight $d(p, q)$ for each ordered pair of distinct permutations.

The overlap graph and the reduction are Houston's [3, Section 2]; what Lemma 2.5 adds is the exact treatment of the inequality gaps.

Lemma 2.5 (Reduction). $M(6) = 6 + C^*$, where C^* is the minimum weight of a Hamiltonian path in the overlap graph. Moreover every superpermutation S , whose length we write $|S|$, induces an ordering $q_1, \dots, q_{720}$ of the permutations, by position of first occurrence, with

$$|S| = 6 + \sum_{k=1}^{719} d(q_k, q_{k+1}) + \text{slack}(S), \quad \text{slack}(S) \geq 0,$$

and conversely gluing any ordering $q_1, \dots, q_{720}$ by maximal overlap yields a superpermutation of length exactly $6 + \sum d$, whose own first-occurrence ordering need not be the one glued.

Proof. Gluing an ordering by maximal overlap gives a superpermutation of length exactly $6 + \sum d$, so $M(6) \leq 6 + C^*$. Conversely, listing the permutations of S by first-occurrence position p_k , the string must spell out enough of q_{k+1} after q_k to complete it, so $p_{k+1} - p_k \geq d(q_k, q_{k+1})$; summing, with the six characters before the first window ends and those after the last, gives the identity with $\text{slack}(S)$ collecting the gaps, and $|S| \geq 6 + C^*$. $\square$

2.2 The padding argument

Lemma 2.6 (Padding). If no superpermutation on $\{1, \dots, 6\}$ has length exactly 871, then none has length at most 871.

Proof. Append $871 - |S|$ arbitrary characters to a superpermutation S of length below 871; every window of S is still a window of the result, which is therefore a superpermutation of length exactly 871. $\square$

2.3 First-occurrence structure

Fix a superpermutation $S = s_1 s_2 \dots s_m$ over $\{1, \dots, 6\}$, so that its windows are the strings $s_i s_{i+1} \dots s_{i+5}$ for $1 \leq i \leq m - 5$, and let $q_1, \dots, q_{720}$ be its first-occurrence ordering, with costs $d_k = d(q_k, q_{k+1})$. Define

$$\text{rot}(p) = p_2 p_3 p_4 p_5 p_6 p_1, \quad \text{sig}(p) = p_3 p_4 p_5 p_6 p_2 p_1, \quad \rho(p) = p_2 p_3 p_4 p_5 p_1 p_6.$$

The *rotation class* of p is $C(p) = \{\text{rot}^i(p)\}$; there are 120 classes of six permutations each, and the weight-1 arcs of the overlap graph are exactly the arcs $p \rightarrow \text{rot}(p)$. The map ρ fixes the last letter and rotates the first five; it has order exactly 5 on every permutation.

Put $\text{waste} = \sum_k (d_k - 1)$, the total cost of the 719 steps above the minimum of one character each. By Lemma 2.5,

$$|S| = 725 + \text{waste} + \text{slack}, \quad \text{slack} \geq 0. \quad (1)$$

A *run* is a maximal block of consecutive q_i joined by cost-1 steps; a run stays inside one rotation class. A *transition* is a step of cost at least 2. Set $e = \# \text{runs} - 120$, the number of *extra* runs; $h = \#\{k : d_k \geq 3\}$, the number of *heavy* transitions; $X4h = \#\{k : d_k \geq 4\}$; and $X4 = \sum_{d_k \geq 4} (d_k - 3)$. For each class C let y_C be its last element in the ordering and $\text{crit}(C) = \text{rot}(y_C)$ its *critical* permutation; each $\text{crit}(C)$ starts a run, so the runs split into 120 *critical* runs and e non-critical ones, and a run end is a *completion* if it finishes its class and *premature* otherwise.

A cost-2 step from z lands on $\text{rot}^2(z)$ or on $\text{sig}(z)$, these being the two ways to complete the last four letters of z . A cost-2 exit from a completion y_C lands on $\text{sig}(y_C)$, since $\text{rot}^2(y_C)$ lies in the finished class C ; if

that landing is the critical permutation of its class the step is a *link*, and otherwise a *non-link*, and cnl is the number of non-links. Links chain the 120 classes into *link-paths*, along which the critical permutations advance by ρ , so a link-path holds at most five classes. A path holding ℓ classes has *deficiency* $5 - \ell$; D is the total deficiency, P the number of paths, S_5 the number of paths of length exactly 4, S_4 the number of length at most 3, and $P_s = S_5 + S_4$ the number of *short* paths, those of length at most 4. A *hop* is a cost-3 exit from the end of one path landing on the head critical permutation of another; hops join paths into *hop-chains*, ch is their number, and the *cost* of a chain is the total deficiency of its members.

A premature cost-2 exit from z is a *re-entry* if it lands on $\text{rot}^2(z)$, and $Q2$ is the number of re-entries; otherwise it lands on $\text{sig}(z)$, and it is a *premature-exit hit* if that landing is critical. Five *defect* counts record departures from the clean pattern: $zq \in \{0, 1\}$ is 1 exactly when q_1 is not critical; zh counts heavy transitions landing on a non-critical permutation; $zp3$ counts premature run ends exiting at cost at least 3; $zp2s$ counts re-entries landing on a non-critical permutation; and $zp2g$ counts premature cost-2 exits landing on a non-critical $\text{sig}(z)$. Their sum is $Z = zq + zh + zp3 + zp2s + zp2g$, and *hits* is the number of premature-exit hits. The *pointer graph* has the runs as vertices and one arc per cost-2 transition other than a re-entry, from the run of the exiting permutation's rotation to the run of the landing; cyc is its number of cycles and $Ncomp$ its number of components.

2.4 The accounting identity

Lemma 2.7 (Accounting identities). *For every first-occurrence ordering on $\{1, \dots, 6\}$, with no hypothesis on the length,*

$$P = 24 + \frac{D}{5}, \quad h = 23 + \frac{D}{5} + Z - e, \quad \text{waste} = 142 + \frac{D}{5} + X4 + Z.$$

Proof. Link-path lengths are at most 5 and sum to 120, giving $P \geq 24$ and $D = 5P - 120$, which is the first identity. For the second, count the $119 + e$ transitions by type: each of the 120 completions except the global last has exactly one exit, which is a link, a non-link, or heavy, and eliminating the typed counts leaves $h = 23 + D/5 + Z - e$. For the third, summing costs gives $\text{waste} = 119 + e + h + X4$, and substituting the second identity gives $\text{waste} = 142 + D/5 + X4 + Z$. $\square$

We call $(D, X4, Z)$ the *type* of an ordering, and a *structural class* is a pair (type, e); because the identity is an equality, the type fixes the waste. Table 1 collects the notation.

Table 1: Notation. Everything is read off the first-occurrence ordering $q_1, \dots, q_{720}$ of a fixed superpermutation.

$d(p, q)$	characters needed after p to reach q ; $\text{rot}, \text{sig}, \rho$ the three maps of Section 2.3
$C(p)$	rotation class of p ; y_C its last element; $\text{crit}(C) = \text{rot}(y_C)$
waste, slack	$\sum_k (d_k - 1)$, and $ S - 725 - \text{waste}$
run, e	maximal cost-1 block; $e = \#\text{runs} - 120$, the non-critical runs
completion, premature	a run end that finishes its class, or does not
$h, X4h, X4$	$\#\{d_k \geq 3\}$, $\#\{d_k \geq 4\}$, $\sum_{d_k \geq 4} (d_k - 3)$
link, non-link, cnl	completion cost-2 exit onto a critical, or a non-critical, permutation; count of the latter
link-path, ℓ , deficiency	maximal chain of links; its length ≤ 5 ; $5 - \ell$
D, P, S_5, S_4, P_s, F_p	total deficiency; number of paths; of length 4; of length ≤ 3 ; short = $S_5 + S_4$; full = $P - P_s$
hop, hop-chain, ch , cost	cost-3 exit from a path end onto a path head; maximal chain of hops; count; total deficiency
singleton, N_s	a hop-chain that is one short path; their number
re-entry, $Q2$	premature cost-2 exit onto rot^2 of the source; count
hit, $hits$	premature cost-2 exit onto a critical sig ; count
$zq, zh, zp3, zp2s, zp2g, Z$	the five defect counts of Section 2.3 and their sum
pointer graph, $cyc, Ncomp$	runs joined by cost-2 arcs other than re-entries; cycles; components
type, B , class	$(D, X4, Z)$; $D/5 + X4 + Z = \text{waste} - 142$; a type with a value of e

3 The zero-slack classification

At length 871 the allowance over the floor in (1) is small enough to be accounted for completely. The analysis below shows that a candidate must spend it exactly, with *zero slack*, and that only sixteen structural classes survive, the *classification* of Theorem 3.10.

Lemma 3.1 (Base bound). *Every first-occurrence ordering on $\{1, \dots, 6\}$ has waste ≥ 143 .*

Proof. By the converse direction of Lemma 2.5, an ordering of waste w yields a superpermutation of length $725 + w$. The bound of [5] gives $M(6) \geq 867$, and Houston’s improvement by one, recorded in the remark following Theorem 2.5 of [4] and proved in [6], gives $M(6) \geq 868$; hence $725 + w \geq 868$, that is $w \geq 143$. $\square$

Remark 3.2. This is the one number taken from the literature; the repository also derives it independently (`src/audit/checks/itemA_scan6.py`). The bound $M(6) \geq 869$ of [7] is not used.

So let Σ be a superpermutation of length exactly 871. By (1), waste + slack = 146, and by Lemma 3.1 and slack ≥ 0 we get waste $\in \{143, \dots, 146\}$. Setting

$$B := \frac{D}{5} + X4 + Z = \text{waste} - 142$$

via Lemma 2.7, we have $B \in \{1, 2, 3, 4\}$ and slack = $4 - B$. There are exactly thirty-four types with $1 \leq B \leq 4$, namely 3, 6, 10, 15 at $B = 1, 2, 3, 4$, and the case analysis runs over them and, inside each, over the values of e permitted by $h \geq 0$.

Lemma 3.3 (Chain capacity). *In every first-occurrence ordering on $\{1, \dots, 6\}$:*

1. *a maximal stretch of consecutive link-paths of lengths in $\{4, 5\}$ within one hop-chain contains at most 4 paths, at most 2 of length 4, and no length-4 path in an interior position;*
2. *for every cost $c \leq 20$ and endpoint type $(a, b) \in \{0, 1\}^2$, where a and b record whether the first and last member paths are short, the maximum number of paths in a hop-chain of that cost and type is a tabulated value $g(c, a, b)$, with $g(20, \cdot) = 25$;*
3. *for every $c \leq 18$, every prescription (s_5, s_4) of how many members have length 4 and how many length at most 3, and every endpoint type, the maximum is a tabulated value g_S ; prescriptions absent from the table are unrealisable;*
4. *a hop-chain with m link-paths, cost c and endpoint type (a, b) satisfies $m - 2c \leq 4 - 2a - 2b - s$, where $s = 1$ if the chain is a single short path and $s = 0$ otherwise.*

The tables, and the four typed maxima of $m - 2c$ behind clause (4), are certified by exhaustive enumeration over *abstract* chains: sequences of pairwise class-disjoint ρ -consecutive segments joined by transitions of cost exactly 3. Every hop-chain of an ordering is such a sequence, since critical permutations advance by ρ along a link-path, links partition the classes among the link-paths, and hops have cost exactly 3; so a maximum over the abstract domain bounds every real chain. Relabelling the alphabet acts transitively on the permutations and commutes with rot , ρ , sig and d , so the head may be normalised to 123456. The enumeration is Section 5.2.

Definition 3.4. Write $\text{cap}(c, a, b) = \max_{c' \leq c} g(c', a, b)$, the envelope of the table, which bounds a chain of cost at most c . For $k, m, sh, st \geq 0$ with $m \leq 20$ let $F(k, m, sh, st)$ be the maximum of $\sum_{i=1}^k \text{cap}(c_i, a_i, b_i)$ over choices of costs $c_i \geq 0$ with $\sum c_i \leq m$ and endpoint types (a_i, b_i) with $\sum a_i \geq sh$ and $\sum b_i \geq st$. Because the objective is a sum of one term per chain, F is the value of the recurrence

$$F(k, m, sh, st) = \max_{\substack{0 \leq c \leq m \\ (a, b) \in \{0, 1\}^2}} \left[\text{cap}(c, a, b) + F(k - 1, m - c, \max(sh - a, 0), \max(st - b, 0)) \right],$$

with $F(0, m, 0, 0) = 0$ and $F(0, m, sh, st) = -\infty$ when $sh + st > 0$; it is non-decreasing in k and m and non-increasing in sh and st .

Every type scanned below has $B \leq 4$, hence $D \leq 20$, so every application of F and of the tables in this paper has $m \leq 20$.

Lemma 3.5 (Necessity of the packing test). *If a set of k hop-chains of an ordering carries B link-paths in total, has total cost at most $m \leq 20$, and contains at least sh chains whose first path is short and at least st whose last path is short, then $B \leq F(k, m, sh, st)$.*

Proof. The real chains realise one of the tuples (c_i, a_i, b_i) over which the maximum is taken, and each chain’s path count is at most $g(c_i, a_i, b_i) \leq \text{cap}(c_i, a_i, b_i)$ by Lemma 3.3(2). $\square$

Three further counting statements are used, each valid for every ordering.

Lemma 3.6 (Singleton-chain bound). *Call a hop-chain a singleton if it consists of one short path, and let N_s be their number. Then $N_s \geq \max(OV, cyc - zp2g)$, where $OV = \max(0, SH + ST - P_s)$, with SH and ST the numbers of chains obliged to begin and to end with a short path, and $zp2g$ is the defect count of premature cost-2 exits landing on another run's start; and, if $D \leq 20$, for every $n_s \leq N_s$, $D \geq n_s$ and $P \leq n_s + F(ch - n_s, D - n_s, 0, 0)$, where ch is the number of chains.*

Proof. Two lower bounds on N_s . The SH chains obliged to begin short and the ST obliged to end short draw their endpoints from the P_s short paths, so at least $SH + ST - P_s$ of those paths serve both roles at once, and such a path is the first and the last member of its chain, hence a singleton. Separately, consider a pointer cycle containing no $zp2g$ arc. An arc with a non-critical head is a non-link or a $zp2g$ arc, and an arc with a non-critical tail, the tail being the run of $\text{rot}(z)$ for the source z , is a premature sig-exit, which lands on a critical permutation (a hit) or is a $zp2g$ arc; so on such a cycle every non-critical run is entered by a non-link and left by a hit, and the cycle alternates non-critical runs with maximal blocks of critical runs, of which there is at least one of each, since arcs between critical runs are links and links advance in first-occurrence order and hence are acyclic; and pointer arcs advance run starts by ρ , whose order is five, so the cycle has at most five vertices. A critical block is entered by a hit, so its first class is not link-entered and heads its link-path, and it is left by a non-link, so its last class ends its link-path; the block is therefore a single link-path, short by the five-vertex bound, and it heads its hop-chain, a hop having cost 3, and ends it. Each such cycle thus yields a singleton, distinct cycles yield distinct ones, and the cycles containing a $zp2g$ arc number at most $zp2g$, the arcs of distinct cycles being distinct. For the second clause, a singleton short chain has one path of cost $5 - \ell \in \{1, \dots, 4\}$, so deleting n_s of them removes n_s chains, n_s paths and at least n_s of the cost budget, and discharges rather than creates endpoint obligations; F is non-decreasing in its cost argument, so Lemma 3.5 applied to the remainder gives the bound. $\square$

Lemma 3.7 (Refined chain bookkeeping). *Let a hop-chain have cost c , s_5 members of length 4, s_4 members of length at most 3, endpoint type (a, b) , r maximal $\{4, 5\}$ -stretches, and let $a_3, b_3 \in \{0, 1\}$ record whether its first, respectively last, member has length at most 3. Then $r \leq s_4 + 1 - a_3 - b_3$, its member count is at most $s_4 + 4r$, and $s_5 \leq 2r$; if $c \leq 18$ its tuple (c, s_5, s_4, a, b) is moreover an entry of g_S and its member count is at most that entry. The per-chain totals sum to D , S_5 and S_4 exactly. The scanner charges a chain of cost at most 18 by the smaller of g_S , these run inequalities and $\text{cap}(c, a, b)$, and a chain of cost 19 or 20 by the smaller of the run inequalities and $\text{cap}(c, a, b)$.*

Proof. Distinct maximal $\{4, 5\}$ -stretches of a chain are separated by at least one member of length at most 3, and a leading or trailing such member is not a separator, which gives the bound on r ; each stretch is itself a chain, so Lemma 3.3(1) caps it at four members with at most two of length 4, which gives the other two inequalities. The g_S clause is Lemma 3.3(3). The hop relation is acyclic, since a hop lands on a later first occurrence, so the link-paths partition into hop-chains and the totals are exact. $\square$

Lemma 3.8 (Component packing). *Every component of the pointer graph carries whole maximal chains of critical runs and has at most 5 vertices; a component carrying m link-paths contains at least $m - 1$ non-critical runs, and m if it is a cycle; and the number of components satisfies the exact identity $N_{\text{comp}} = P - \text{cnl} + Q2 + zp3 + \text{cyc}$. Consequently the multiset of link-path lengths must pack into the components under these constraints.*

Proof. Pointer in- and out-degrees are at most 1, so components are directed paths or cycles, and every arc advances the run start by ρ , of order 5, so a component has at most 5 vertices. Links are pointer arcs, so the critical runs of a link-path lie in one component, which is the whole-chains clause. A path-head critical run can be entered inside its component only by a premature-exit hit whose tail is a non-critical run, since a link would not leave it heading a path, a non-link lands on a non-critical run, and a re-entry generates no arc; distinct heads consume distinct such tails, and in a cycle every vertex has an in-arc while in a path all but one do. Finally $V = 120 + e$ and $E = (120 - P) + \text{cnl} + (e - Q2 - zp3)$, counting links, non-links and premature cost-2 arcs, so $N_{\text{comp}} = V - E + \text{cyc}$ is as stated. $\square$

A *profile* is a tuple of the counts $(D, X4, Z, e)$, the five defect counts, $Q2$, cyc , S_5 , S_4 , $X4h$, ch and the slack; its derived quantities are

$$P = 24 + \frac{D}{5}, \quad h = 23 + \frac{D}{5} + Z - e, \quad cnl = e - (zq + zh + zp2s + zp2g), \quad hits = e - Q2 - zp3 - zp2g,$$

$F_p = P - P_s$, $SH = hits$, $ST = cnl$, $OV = \max(0, SH + ST - P_s)$, $PEN = \max(OV, cyc - zp2g)$ and $R_{cap} = 1 + X4h + zh + cnl + S_4$. The next lemma is the complete list of what the scanner tests.

Lemma 3.9 (Scanner inequalities). *Every first-occurrence ordering on $\{1, \dots, 6\}$ with $D \leq 20$, of any length, satisfies:*

- (S1) $h \geq 0$, where h is the number of heavy transitions;
- (S2) $cnl \geq 0$, where cnl is the number of non-links;
- (S3) $hits \geq 0$, where $hits$ is the number of premature-exit hits;
- (S4) $Q2 \leq zp2s + \lfloor \text{slack} / 4 \rfloor$;
- (S5) $cyc \leq e - Q2$;
- (S6) if $Z = 0$ then $D \geq e + 5(e - Q2 - cyc)$;
- (S7) $\max(0, D - 4S_4) \leq S_5 \leq D - 2S_4$, $P_s \leq D$, and $P_s = 0$ if $D = 0$;
- (S8) $hits \leq P_s$ and $cnl \leq P_s$;
- (S9) $ch \leq 1 + X4h + zh + cnl$, $hits \leq ch$ and $cnl \leq ch$;
- (S10) $F_p + S_5 \leq 4R_{cap}$ and $S_5 \leq 2R_{cap}$;
- (S11) the number N_s of singleton chains satisfies $N_s \geq PEN$; in particular $PEN \leq ch$;
- (S12) $P - 2D \leq 4ch - 2(SH + ST) - PEN$;
- (S13) $P \leq PEN + F(ch - PEN, D - PEN, \max(0, SH - PEN), \max(0, ST - PEN))$, and P is at most the refined maximum of Lemma 3.7;
- (S14) the multiset of link-path lengths packs into the pointer components as in Lemma 3.8.

In (S1)–(S3) the content is that the displayed expressions equal the named counts. The scanner evaluates (S9) and (S10) with $X4h$ replaced by $X4 \geq X4h$, and it evaluates SH and ST as $hits$ and cnl , which are lower bounds on the numbers of chains whose first, respectively last, path is short; each substitution weakens the test it enters, so every profile the scanner rejects violates the lemma.

Proof. Four facts are used throughout. (a) $\text{crit}(C)$ starts a run, since otherwise its predecessor in the run would be y_C ; there are $120 + e$ runs, 120 completions and e premature ends, and the run starts are the 120 critical permutations and e non-critical ones. (b) Every cost-2 landing, and $\text{rot}(z)$ for every premature run end z , starts a run, and every non-critical run start other than q_1 is the landing of exactly one transition. (c) Along a link the critical permutation advances by ρ , because $\text{sig}(\text{rot}^{-1}(x)) = \rho(x)$; the link relation and the hop relation are acyclic, since a link or hop lands on a permutation whose first occurrence precedes the end of the landing class; so the classes partition into P link-paths of length at most 5 with $D = 5P - 120$, the paths partition into hop-chains, and the links number $120 - P$. (d) Every pointer arc advances the run start by ρ , so, ρ having order 5 and degrees being at most 1, pointer components are directed paths or cycles with at most five vertices.

(S1)–(S3). Of the 120 completions all but q_{720} have an exit, a link, a non-link or a heavy transition; by (c) the links number $120 - P$, so the heavy completion exits number $P - 1 - cnl$, and adding the $zp3$ premature heavy exits and substituting $P = 24 + D/5$ and $cnl = e - (Z - zp3)$ gives $h = 23 + D/5 + Z - e$. Each of the e non-critical run starts is q_1 , when $zq = 1$, or the landing of a transition onto a non-critical permutation, which is a non-link, a re-entry, a premature sig-exit or a heavy transition, counted by cnl , $zp2s$, $zp2g$ and zh ; this is (S2). Each of the e premature run ends exits once, as a re-entry, a heavy transition, or a sig-exit landing on a non-critical or on a critical permutation; this is (S3).

(S4). Write the first occurrences at positions $p_1 < \dots < p_{720}$ with gaps $g_k = p_{k+1} - p_k \geq d_k$. Suppose $d_k < g_k < 6$, and let u be the last $6 - d_k$ letters of q_k , which by the definition of d_k are also the first $6 - d_k$ letters of q_{k+1} . Then u occurs in the string at position $p_k + d_k$ and again at position $p_k + g_k$, and the two occurrences overlap because their offset $t = g_k - d_k$ is less than $|u| = 6 - d_k$; comparing the string letter by letter on the overlap gives $u_i = u_{i+t}$ for $1 \leq i \leq |u| - t$, so two distinct positions of u carry the same letter, which is impossible since u is part of a permutation. Hence $g_k = d_k$ or $g_k \geq 6$, that is $g_k - d_k \geq 6 - d_k$. Let $z = q_k$ be the source of a re-entry, so $d_k = 2$. Either $g_k - d_k \geq 4$, and this re-entry alone consumes four units of slack, or $g_k = 2$, in which case the window at $p_k + 1$ is $\text{rot}(z)$, which is not a first occurrence and

so occurs before z ; then $\text{rot}(z) \neq y_{C(z)}$, the landing $\text{rot}^2(z)$ is not critical, and the re-entry is a $zp2s$ unit. Hence $Q2 - zp2s \leq \lfloor \text{slack} / 4 \rfloor$.

(S5). For a premature cost-2 source z the run starting at $\text{rot}(z)$ is non-critical, by (b) and the choice of $y_{C(z)}$, and distinct z give distinct such runs. A re-entry generates no pointer arc, and its run at $\text{rot}(z)$ is the tail of no other arc, so that run has out-degree 0 and lies on no cycle. Every pointer cycle contains a non-critical run: if it contains a critical run it contains the whole critical chain of that run's link-path, and the arc leaving the last critical run of the chain is a completion cost-2 exit that is not a link, hence a non-link with a non-critical head. Cycles are disjoint, so $Q2 + cyc \leq e$.

(S6). At $Z = 0$, $cnl = e$ and every non-critical run is entered by a non-link, whose tail is a critical run, so every pointer component contains a critical run. A component with $m_g \geq 1$ link-paths and k_g non-critical runs has at most five vertices by (d), so its deficiency is at least $5m_g - (5 - k_g) = k_g + 5(m_g - 1)$. With $V = 120 + e$ vertices and $E = (120 - P) + e + (e - Q2)$ arcs, $Ncomp = P - e + Q2 + cyc$, and summing over components gives $D \geq e + 5(P - Ncomp)$.

(S7). Each of the S_5 paths of length 4 has deficiency 1 and each of the S_4 paths of length at most 3 has deficiency in $\{2, 3, 4\}$, so $2S_4 \leq D - S_5 \leq 4S_4$; hence $P_s \leq S_5 + (D - S_5)/2 \leq D$, and $D = 0$ forces every path full.

(S8). A premature-exit hit lands on a critical permutation w that, being entered by a premature step, is not link-entered, so w heads its link-path; that path is short, since the four links of a full path already form a five-vertex pointer component and the hit's arc would add a sixth vertex, against (d). Distinct hits give distinct heads. Dually a non-link leaves a class with no outgoing link, which ends its path, and that path is short by the same five-vertex argument at the other end.

(S9). The head critical permutation of a hop-chain is not hop-entered and is entered, unless it is q_1 , by exactly one transition, which is a transition of cost at least 4, a premature cost-3 exit landing on a critical permutation, a premature-exit hit, or a re-entry landing on a critical permutation; a completion cost-3 exit landing there would be a hop, and a link is excluded. Summing the bounds $1 - zq$, $X4h$, $zp3$ less the premature heavies landing off-critical, $hits$ and $Q2 - zp2s$, and substituting (S2) and (S3), gives $ch \leq 1 + X4h + zh + cnl$. By (S8) a hit-headed path heads its chain, a hop having cost 3, and a non-link-ended path ends its chain, so $hits \leq ch$ and $cnl \leq ch$.

(S10). Let R be the number of maximal stretches of consecutive link-paths of length in $\{4, 5\}$ inside hop-chains. A contiguous piece of a chain is a chain, so Lemma 3.3(1) gives $F_p + S_5 \leq 4R$ and $S_5 \leq 2R$. A stretch ends at the global end of the ordering, at a transition of cost at least 4, at a heavy transition landing off-critical, at a non-link, or at a hop onto the head of a path of length at most 3, and each terminator ends at most one stretch, so $R \leq R_{cap}$.

(S11) is Lemma 3.6. (S12). Give a chain with m paths and cost c the value $m - 2c$. By Lemma 3.3(4) its value is at most $4 - 2a - 2b - s$, and summing over the chains, which partition the paths by (c), gives $P - 2D \leq 4ch - 2SH^* - 2ST^* - N_s$, with SH^* and ST^* the true numbers of chains with short first, respectively last, path. By (S8) these are at least $hits$ and cnl , and $N_s \geq PEN$ by (S11), all three entering with a negative sign.

(S13). Each singleton chain has endpoint type $(1, 1)$, one path and cost in $\{1, \dots, 4\}$, so deleting PEN of them leaves $ch - PEN$ chains carrying $P - PEN$ paths at total cost at most $D - PEN$, with at least $\max(0, SH - PEN)$ obliged to begin short and at least $\max(0, ST - PEN)$ obliged to end short; the first bound is Lemma 3.5 applied to them, F being non-decreasing in its cost argument, and the second is Lemma 3.7. (S14) is Lemma 3.8.

Finally the substitutions. $X4h$ enters (S9) and (S10) only on the right of a bound with coefficient $+1$, so replacing it by $X4 \geq X4h$ weakens both. SH and ST enter (S9) on the left, (S11) and (S12) through OV and with a negative sign, and (S13) as obligations; replacing the true counts by the smaller $hits$ and cnl weakens each occurrence. $\square$

Theorem 3.10 (Classification). *Every superpermutation on $\{1, \dots, 6\}$ of length 871 has waste = 146 and slack = 0, and its structural class is one of exactly sixteen: with the type written $(D, X4, Z)$,*

<i>type</i>	<i>e</i>	<i>classes</i>
(20, 0, 0)	1, 2, 3, 4, 5, 6, 7, 8, 9	9
(15, 1, 0)	1, 2, 3, 4, 7	5
(10, 2, 0)	2	1
(15, 0, 1)	2	1
<i>total</i>		16

Proof. The scan is Section 5.3. For each of the thirty-four types and each admitted e it enumerates every profile and rejects it on any of the fourteen tests (S1)–(S14) of Lemma 3.9, which applies because every scanned type has $B \leq 4$ and hence $D \leq 20$. Each test is valid for every such ordering, so the scan only ever rejects profiles that no ordering realises, and a class survives only if no test bites. Exhausting the enumeration therefore proves that the surviving classes are the only possible ones. Every type with $B \leq 3$, all nineteen of them, is eliminated, so $B = 4$, that is waste = 146 and slack = 0; among the fifteen types with $B = 4$ exactly four survive, with the e -ranges above plus $e = 0$ at (10, 2, 0) and at (15, 1, 0), giving eighteen classes. Those two are removed by Lemma 3.11, leaving sixteen. $\square$

Lemma 3.11 (No unbroken candidate). *No superpermutation of length 871 has $e = 0$.*

Proof. Cut the ordering at its h heavy transitions into $r = h + 1$ segments. Summing costs as in Lemma 2.7, waste = 119 + e + h + $X4$, so the maximal gluing of the ordering has length 843 + e + r + $X4$. The scan above, before invoking the present lemma, has already shown that every surviving candidate has $B = 4$, hence slack = 0; so the candidate is the maximal gluing of its own ordering and $r + X4 = 28$, with $X4 \leq 4$; the searches cover $0 \leq X4 \leq 8$, that is $20 \leq r \leq 28$. The searcher is Section 5.4; all nine instances terminate with the exhaustion marker and no solution. $\square$

Two refinements are needed later; both are carried out by the programs of Section 5.5.

Lemma 3.12 (No very heavy transition). *In every class of Theorem 3.10, $X4h = X4$; in particular no transition of cost at least 5 occurs in any length-871 superpermutation.*

Proof. Since $d \leq 6$, each transition of cost at least 4 contributes between 1 and 3 to $X4$, so $\lceil X4/3 \rceil \leq X4h \leq X4$, with equality on the right exactly when every such transition has cost exactly 4. At (20, 0, 0) and (15, 0, 1) we have $X4 = 0$, forcing $X4h = 0$; at (15, 1, 0), $X4 = 1$ forces $X4h = 1$. Only the cell (10, 2, 0), $e = 2$, $X4h = 1$ remains, where $P = 26$, $D = 10$ and, by Lemma 4.1(1), $ch = 4$. The identity $a_2 + 2a_3 + c_1 + 2c_2 = 2$ of (2) has five solutions, and the deficiency floors of Section 4, at least 6 per two-block component, 12 per three-block one, 1 per type-I and 7 per type-II cycle, dispose of two at once:

a_2	a_3	c_1	c_2	a_1	floor	verdict
0	0	2	0	24	2	continue
0	0	0	1	24	7	continue
0	1	0	0	23	12	empty, floor exceeds D
1	0	1	0	23	7	continue
2	0	0	0	22	12	empty, floor exceeds D

For the three survivors evaluate (3)–(4) at every admissible b_{q_1} :

a_2	a_3	c_1	c_2	b_{q_1}	λ	N_{lo}	N_{hi}	verdict
0	0	2	0	0	2	20	20	continue, $n_{45} = 20$ forced
0	0	0	1	0	2	23	20	empty
1	0	1	0	0	3	22	20	empty
1	0	1	0	1	2	22	20	empty

In the one remaining configuration the link-path lengths sum to 120: the 20 one-block paths of length 4 or 5 and the two cycle paths of length 4 account for $108 - n_4$, leaving $12 + n_4$ for the $n_S = 4$ short one-block paths, each of length at most 3, so $n_4 = 0$. The four chains are the two cycle singletons, of cost 1 each, and two others, which must carry the remaining 24 paths at total cost 8; but $F(2, 8, 0, 0) = \max_c (\text{cap}(c, 0, 0) + \text{cap}(8 - c, 0, 0)) = 20$, so Lemma 3.5 fails and the cell is empty. $\square$

Lemma 3.13 (The defect-one class). *In the class $(15, 0, 1)$, $e = 2$, the single defect unit is a heavy transition landing on a non-critical permutation or a premature cost-3 exit; exactly two component wirings survive, each with its complete skeleton determined, and the ordering has no pointer cycles and exactly three hop-chains.*

Proof. Here $P = 27$, $D = 15$, $X4 = 0$ so $X4h = 0$, and $h = 25$. The single defect unit lies in exactly one of the five pools. Each pool fixes $Q2$, cnl and $hits$ by (S2)–(S4) of Lemma 3.9, since at zero slack $Q2 = zp2s$; and it fixes ch , because the 25 heavy transitions are $zh + zp3$ non-hops and $25 - zh - zp3$ hops, so $ch = P - \#hops = 2 + zh + zp3$. Lemma 3.5, applied to all chains with the obligations $sh = hits$ and $st = cnl$ of (S8), eliminates three pools:

pool	$Q2$	cnl	$hits$	ch	test	verdict
zq	0	1	2	2	$27 \leq F(2, 15, 2, 1) = 24$	empty
zh	0	1	2	3	wiring analysis	below
$zp3$	0	2	1	3	wiring analysis	below
$zp2s$	1	1	1	2	$27 \leq F(2, 15, 1, 1) = 25$	empty
$zp2g$	0	1	1	2	$27 \leq F(2, 15, 1, 1) = 25$	empty

In the two surviving pools there are two non-critical runs and three chains. In the zh pool the run nc_H entered by the defect heavy transition has no pointer in-arc, and both premature ends are hits, so both non-critical runs have an out-arc; in the $zp3$ pool both non-critical runs are entered by non-links and the run nc_2 whose premature end is the defect exit has no out-arc. This in-and-out census, the alternation of critical blocks and non-critical runs, and the five-vertex cap of Lemma 3.8 leave exactly three component shapes per pool, with K_i a block of critical runs and every other run a one-block component: in the zh pool $[nc_H K_1] + [K_2 nc_2 K_3]$, $[nc_H K_1] + cyc[nc_2 K_2]$ and $[nc_H K_1 nc_2 K_2]$; in the $zp3$ pool $[K_1 nc_1 K_2] + [K_3 nc_2]$, $cyc[nc_1 K_1] + [K_2 nc_2]$ and $[K_1 nc_1 K_2 nc_2]$. Each shape is tested in generic form and under every designation of a block as the path of q_1 , the emitter or target of the defect transition, or the global end, which forces that block's path to be a singleton chain; the test is $blocks \leq F$ of Lemma 3.5 over the chains that are not forced singletons, with the cost and the obligations of the forced singletons removed. The generic first shape of each pool passes, with $27 \leq F(3, 15, 2, 1) = 30$ and $27 \leq F(3, 15, 1, 2) = 30$; the other twenty-two rows fail, the generic cycle shapes with $26 > F(2, 14, 1, 0) = 25$ and $26 > F(2, 14, 0, 1) = 25$ and every designated row by a margin of at least two. The full table is printed by the program of Section 5.5. Exactly two wirings survive, one per pool, both cycle-free; a designation only adds a constraint, so no designation of a surviving row is a third wiring. In both the ordering has three hop-chains and no pointer cycle. $\square$

4 Emptiness of the sixteen classes

Showing all sixteen classes empty proves the Main Theorem, by Theorem 3.10. Two separate elimination routes were run. The first eliminates, by capacity bounds and an exact-cover obstruction, the six classes whose direct search would be computationally prohibitive; the second eliminates all sixteen by compiling each into a finite placement problem and searching it to exhaustion, which needs a theorem saying that no candidate escapes the problems so compiled.

Fifteen of the sixteen classes have $Z = 0$, and slack = 0 forces the re-entry count to vanish there too; the remaining class is controlled by Lemma 3.13. Except where that defect class is treated explicitly, we work at $Z = 0$.

4.1 Rigid structure at zero slack

Lemma 4.1. *At $Z = 0$:*

1. $ch = e + 1 + X4h$ exactly; every cost-3 heavy transition is a hop, every heavier one enters the head of a chain, and the chain heads are exactly the path of q_1 , the e paths entered by premature-exit hits, and the $X4h$ paths entered by cost- ≥ 4 transitions;
2. every non-critical run has exactly one incoming pointer arc, a non-link, and one outgoing arc, a premature-exit hit onto a critical run; hence every pointer component alternates maximal blocks of critical runs with single non-critical runs, a path component with m blocks having exactly $m - 1$ non-critical runs;

3. every pointer cycle has exactly five vertices, so it is either one link-path of length 4 with one non-critical run (type I) or two link-paths of lengths $\{1, 2\}$ with two non-critical runs (type II), and every link-path in a cycle is a singleton hop-chain.

Proof. (1) At $Z = 0$ every heavy transition leaves a completion and lands on a critical permutation heading its link-path; for cost 3 that is the definition of a hop. Counting chain heads by their unique entering transition and using $h = P - 1 - e$ from Lemma 2.7 at $Z = 0$ gives the identity. (2) At $Z = 0$ the only arc that can enter a non-critical run is a non-link and the only one that can leave it is a premature-exit hit, and the e of each are in bijection with the e non-critical runs. (3) Pointer arcs advance run starts by ρ , so a j -vertex cycle forces ρ^j to fix a permutation; ρ has order exactly 5, so $j = 5$, and the two shapes follow from (2). A cycle link-path is both entered by a hit and the emitter of a completion non-link, so it both heads and ends its chain. $\square$

Lemma 4.2 (Endpoint lengths). *Every non-cycle hop-chain that is headed by a premature-exit hit begins with a link-path of length at most 3, and every non-cycle chain ended by a completion non-link ends with one.*

Proof. By Lemma 4.1(2) the first block of such a chain is preceded in its pointer component by the non-critical run supplying the hit and, by the alternation, by a further block of at least one vertex; the five-vertex cap then gives $\ell + 1 + 1 \leq 5$. Symmetrically for the last block. $\square$

Write a_m for the number of pointer path components carrying m link-paths and c_m for the number of cycle components carrying m , so c_1 and c_2 count type-I and type-II cycles. Counting non-critical runs and link-paths by component gives

$$a_2 + 2a_3 + c_1 + 2c_2 = e, \quad a_1 + a_2 + a_3 = P - e. \quad (2)$$

The five-vertex cap pins the *deficiency floors*: at least 6 per two-block path component, exactly 12 per three-block one, exactly 1 per type-I cycle and exactly 7 per type-II cycle. Writing n_{45} for the number of one-block components whose path has length 4 or 5, $n_S = a_1 - n_{45}$ for those with a path of length at most 3, and $\lambda = a_2 + a_3 + 1 + X4h - b_{q_1}$ for the number of chains that are not forced singletons (with $b_{q_1} = 1$ exactly when q_1 's component carries at least two link-paths), two consequences are used constantly. The first is the *class-length budget*: link-path lengths sum to 120 and are capped componentwise, so

$$n_{45} \geq N_{\text{lo}} := \left\lfloor \frac{120 - 3a_1 - 4a_2 - 3a_3 - 4c_1 - 3c_2}{2} \right\rfloor. \quad (3)$$

The second is the *run cap*: by Lemma 3.3(1) the maximal $\{4, 5\}$ -stretches inside chains number at most $n_S + \lambda$ and hold at most 4 paths each, so

$$n_{45} \leq N_{\text{hi}} := \left\lfloor \frac{4(a_1 + \lambda)}{5} \right\rfloor. \quad (4)$$

A configuration with $N_{\text{lo}} > N_{\text{hi}}$ admits no value of n_{45} and is realised by no ordering.

Lemma 4.3 (Cycle-shape exclusion). *In every one of the sixteen classes of Theorem 3.10, every component configuration satisfying the constraints above contains no type-II cycle and no three-block path component.*

Proof. For the defect class this is Lemma 3.13, which gives no cycles and only one- and two-block path components. For the fifteen $Z = 0$ classes, and additionally for the cell $(10, 2, 0)$, $e = 2$, $X4h = 1$, so that the argument does not presuppose Lemma 3.12, the finite set examined is the set of tuples (a_2, a_3, c_1, c_2) of non-negative integers with $a_2 + 2a_3 + c_1 + 2c_2 = e$ and $a_1 = P - e - a_2 - a_3 \geq 0$, restricted to $a_3 \geq 1$ or $c_2 \geq 1$. Over the sixteen cells there are 241 such configurations, 150 with $c_2 \geq 1$ and 150 with $a_3 \geq 1$, overlapping in 59. Two tests are applied. The first is $N_{\text{lo}} > N_{\text{hi}}$ from (3)–(4), evaluated at $b_{q_1} = 0$, which is the weakest setting since λ and hence N_{hi} decrease with b_{q_1} ; it needs no table and eliminates 229 of the 241, including all 59 in the overlap. Representative rows:

cell	a_2	a_3	c_1	c_2	a_1	λ	N_{lo}	N_{hi}
$(20, 0, 0)$ $e=3$	1	0	0	1	24	2	21	20
$(15, 1, 0)$ $e=7$	5	0	0	1	15	7	26	17
$(10, 2, 0)$ $e=2, X4h=1$	0	0	0	1	24	2	23	20

The second test is Lemma 3.5 applied to the λ chains that are not forced singletons, in the form (C11) of Lemma 4.6 below, whose inputs are functions of $(a_2, a_3, c_1, c_2, b_{q_1})$ alone; it eliminates the remaining 12 configurations, each on one entry of the capacity table, at every admissible b_{q_1} . Representative rows, with blocks and cost the paths and the deficiency left to the λ chains after the forced singletons are removed:

cell		a_2	a_3	c_1	c_2	b_{q_1}	blocks	λ	cost	sh	st	F
(20,0,0)	$e=2$	0	0	0	1	0	26	1	13	0	0	19
(20,0,0)	$e=4$	1	0	1	1	1	24	1	10	1	0	15
(20,0,0)	$e=2$	0	1	0	0	0	27	2	16	1	1	26

No configuration survives; the complete lists of 229 and 12 rows are printed by the program of Section 5.5. $\square$

4.2 First route: counting and covering

The first route avoids the full placement search. It eliminates the six classes (20,0,0) with $e \in \{5, \dots, 9\}$ and (15,1,0) with $e = 7$ by counting link-paths against the capacity tables and by an exact-cover test on the rotation classes the cycles would have to occupy.

Lemma 4.4 (Covering obstruction). *At $c_2 = 0$ the classes covered by a type-I cycle's link-path form a ρ -segment $\{C(x), C(\rho x), C(\rho^2 x), C(\rho^3 x)\}$, and the cycles and the non-cycle chains have disjoint class sets partitioning the 120 classes; so once the non-cycle chains are fixed, the leftover classes must admit a partition into one ρ -segment of length four per cycle.*

Proof. A type-I cycle carries one link-path of length four, whose classes form a ρ -segment since critical permutations advance by ρ along it; links partition the 120 classes into link-paths, and grouping by chain gives the statement. $\square$

Proposition 4.5. *The six classes (20,0,0), $e \in \{5, \dots, 9\}$, and (15,1,0), $e = 7$, are empty.*

Proof. Under the constraints of Section 3 and Lemma 4.1 the six classes admit fourteen profiles (cyc, S_5, S_4, ch). Two of them need a pointer cycle of a shape other than the two of Lemma 4.1(3) and are excluded outright. For each of the other twelve, every split of the e non-critical runs into c_1 type-I cycles and $\kappa = e - c_1$ runs in path components ($c_2 = 0$ by Lemma 4.3), every distribution of the S_5 and S_4 short paths among the non-cycle chains, and the endpoint lengths pinned by Lemma 4.2 give a chain prescription, which was put to the exact enumerator of Section 5.6; a profile is empty when every prescription that would realise it is answered negatively. Writing a profile as (cyc, S_5, S_4, ch), the chain queries empty (4,5,6,6), (4,6,6,6) and (4,4,7,6) at (20,0,0), $e = 5$; (5,7,5,7), (5,8,5,7) and (5,6,6,7) at $e = 6$; (6,8,5,8) at $e = 7$; (7,8,5,9) at $e = 8$; and (8,10,4,10) at $e = 9$. The three profiles that survive the chain queries, (6,6,6,7) at $e = 6$, (6,6,6,8) at $e = 7$ and (7,7,4,9) at (15,1,0), $e = 7$, have $\kappa = 0$; their chain decompositions are unique up to composition, and for each the leftover rotation classes were put to the exact-cover test of Lemma 4.4, 59 configurations over the four decisive runs, with no partition found. $\square$

4.3 Second route: the geometric search

The 720 permutations fall into 144 orbits of ρ , and Lemma 4.1 puts each piece of a candidate's structure into consecutive positions of a single orbit; a candidate is therefore a layout of shapes on 144 tracks of five slots, and deciding a class is deciding whether such a layout exists. The route has three ingredients: a census of configurations per class, a placement model proved to over-cover every candidate, and a complete search of every instance.

The census. For each $Z = 0$ class, the generator of Section 5.7 enumerates all values of the configuration tuple $(a_2, a_3, c_1, c_2, b_{q_1}, n_{45}, n_4, \Sigma_{c_1}, \Sigma_{c_2})$ over ranges forced by (2), applying the eleven rejection tests (C1)–(C11) of Lemma 4.6 and emitting everything it does not reject. Each is a necessary condition on a real ordering, so the output over-approximates the realisable set and exhausting it loses nothing. It emits 200 configurations over the fifteen $Z = 0$ classes; the defect class contributes the two wirings of Lemma 3.13 instead; expanding each into concrete piece multisets, with no test applied, yields 631 search instances.

Lemma 4.6 (Census tests). *Let Σ be a superpermutation of length 871 in a class with $Z = 0$, so that $\text{slack} = 0$, $Q2 = 0$ and Lemma 4.1 applies, and read the configuration counts off its ordering. Write n_5 and n_4 for the numbers of one-block components whose path has length 5, respectively 4, so $n_{45} = n_5 + n_4$ and $n_S = a_1 - n_{45}$; write Σ_{c_1} and Σ_{c_2} for the total link-path length carried by the type-I and by the type-II cycles; and put $\text{rem} = 120 - 5n_5 - 4n_4 - 3a_3 - \Sigma_{c_1} - \Sigma_{c_2}$, $sh = a_2 + a_3$, $st = \max(0, a_2 + a_3 - b_{q_1})$, $\text{blocks}_{\text{rest}} = a_1 + 2a_2 + 2a_3 - b_{q_1}$ and $\text{cost}_{\text{rest}} = D - c_1 - 7c_2 - 4a_3 - 2b_{q_1}$. Then:*

- (C1) $c_1 = e - a_2 - 2a_3 - 2c_2 \geq 0$;
- (C2) $a_1 = P - e - a_2 - a_3 \geq 0$;
- (C3) $b_{q_1} = 0$ unless $a_2 + a_3 \geq 1$;
- (C4) $\Sigma_{c_1} = 4c_1$ and $\Sigma_{c_2} = 3c_2$, and every three-block component carries three paths of length 1;
- (C5) $n_{45} \leq 4(n_S + \lambda)$;
- (C6) $n_4 \leq 2(n_S + \lambda)$;
- (C7) $\text{rem} \geq n_S + 2a_2$;
- (C8) $\text{rem} \leq 3n_S + 4a_2$;
- (C9) $P - 2D \leq 4\lambda - 2sh - 2st - c_1 - 12c_2 - 7a_3 - 3b_{q_1}$;
- (C10) $\text{cost}_{\text{rest}} \geq 0$;
- (C11) $\text{blocks}_{\text{rest}} \leq F(\lambda, \text{cost}_{\text{rest}}, sh, st)$.

Proof. (C1) and (C2) are (2). (C3): $b_{q_1} = 1$ means that the component of q_1 carries at least two link-paths, and it is a path component, since the run of q_1 has no incoming arc; so it is counted by a_2 or a_3 . (C4) is Lemma 4.1(3) for the cycles, and for a three-block component the five-vertex cap with two non-critical runs leaves three vertices for three paths.

For the rest, call a chain a *forced singleton* if it is a cycle path, the middle path of a three-block component, or the path of q_1 when $b_{q_1} = 1$. A cycle path is a singleton chain by Lemma 4.1(3); the middle path is entered by a hit from the non-critical run before it and exits by a non-link to the one after it, so it heads and ends its chain; the path of q_1 heads its chain as the global start and, when its component has a second block, exits by a non-link. These $c_1 + 2c_2 + a_3 + b_{q_1}$ chains are distinct, and by Lemma 4.1(1) and (2) the other chains number $ch - (c_1 + 2c_2 + a_3 + b_{q_1}) = \lambda$ and carry $\text{blocks}_{\text{rest}}$ paths.

(C5), (C6). A path of length 4 or 5 in one of the λ chains belongs to a one-block component, since cycle paths are singletons and a path in a multi-block component has length at most 3 by the five-vertex cap; so n_{45} counts these paths and n_4 those of length 4. Inside a chain consecutive such paths are separated by short one-block paths, so the λ chains contain at most $n_S + \lambda$ maximal $\{4, 5\}$ -stretches, and Lemma 3.3(1) bounds each stretch by four paths, at most two of them of length 4. (C7), (C8). The link-path lengths sum to 120, each short one-block path has length in $\{1, 2, 3\}$, and each two-block component has total length in $\{2, 3, 4\}$; subtracting the lengths fixed by (C4), n_5 and n_4 leaves rem for the n_S short one-block paths and the a_2 two-block components.

(C9). Give a chain with m paths and cost c the value $m - 2c$, so that $P - 2D$ is the sum of the values over the chains, which partition the paths. A type-I cycle path has value -1 ; the two paths of a type-II cycle have values -7 and -5 ; the middle path of a three-block component has value -7 ; and the path of q_1 , when $b_{q_1} = 1$, has length at most 3 and value at most -3 . Each of the λ other chains has value at most $4 - 2a - 2b$ by Lemma 3.3(4), and at least sh of them begin with a short path, namely those headed by the second path of a two-block or the third path of a three-block component, which are hit-headed and hence begin short by Lemma 4.2, while at least st end with one, namely those ended by the first path of a multi-block component other than the forced singleton of q_1 . (C10), (C11). The forced singletons have cost exactly 1 per type-I cycle, 7 per type-II cycle and 4 for the middle path of a three-block component, and at least 2 for the path of q_1 , so the λ other chains have total cost at most $\text{cost}_{\text{rest}}$, which is therefore non-negative, and Lemma 3.5 applied to them with the obligations just counted gives (C11), F being non-decreasing in its cost argument. $\square$

The model. Fix an instance. After normalising by a relabelling, a candidate lays down one *arc* of consecutive positions per one-block component, one *span* (block, gap, block) per two-block component, and one full orbit per cycle, pairwise position-disjoint and covering each rotation class once; chains are joined through cost-3 hops, each with exactly six possible targets, and through cost-4 joins where the type demands them, cost 4 being the only possibility by Lemma 3.12; a final exact cover places the cycles, all type I by Lemma 4.3. Every constraint is a proved property of real candidates, so the model over-covers.

Lemma 4.7 (Closure topology). *Let CG be the directed graph whose vertices are the hop-chains not lying in pointer cycles and whose arcs record that the end of one chain triggers the head of another, by closing a multi-block path component, by a cost- ≥ 4 join, or by the defect join in the class of Lemma 3.13. Then CG is one directed path, from the chain of q_1 to the chain containing the global end, together with vertex-disjoint directed cycles; every such cycle contains one of the $a_2 + 2a_3$ arcs routed through a non-critical run of a path component or, in the defect class, the defect arc, so the number of cycles is at most U , where $U := a_2 + 2a_3$ at $Z = 0$ and $U := a_2 + 2a_3 + 1$ in the defect class; and $U \leq 2$ in every one of the sixteen classes.*

Proof. First suppose $Z = 0$, so that Lemma 4.1 applies. Each of the e non-critical runs receives one non-link and generates one hit, giving one candidate arc; of those runs $c_1 + 2c_2$ lie in cycles, where both endpoints are cycle chains and hence not vertices of CG , and the remaining $a_2 + 2a_3$ have both endpoints free. Each of the $X4h$ heavy joins likewise runs between free chains. So $|E(CG)| = a_2 + 2a_3 + X4h$ and, by Lemma 4.1(1) and (2), $|V(CG)| = ch - (c_1 + 2c_2) = a_2 + 2a_3 + 1 + X4h$. In- and out-degrees are at most 1, and the unique source and sink are q_1 's chain and the global end's, so a partial-functional digraph with $|E| = |V| - 1$ is one path plus vertex-disjoint cycles. For the second clause write $t(A)$ for the position of the completion of the last class of chain A ; the completions of a chain occur in increasing order, since each link or hop lands on the critical permutation of the next class, which precedes that class's completion. A cost- ≥ 4 join from A to B is a single transition: it leaves the completion at position $t(A)$ and lands at position $t(A) + 1$ on the head critical permutation of B , which precedes the completion of B 's head class, so $t(A) < t(B)$. A cycle of such arcs alone would give $t(A) < t(A)$. No such monotonicity holds for an arc through a non-critical run, whose outgoing hit is emitted not by the run's own premature end but by the permutation whose rotation starts the run, which may occur anywhere in the ordering; that is why closure loops are possible at all. For the third, grouping D by component and using the deficiency floors gives $D \geq \Delta_1 + 6a_2 + 12a_3 + c_1 + 7c_2$, and eliminating c_1 by (2) gives

$$5\Lambda \leq D - e - \Delta_1 \leq D - e, \quad \Lambda := a_2 + 2a_3 + c_2 \geq U, \quad (5)$$

with Δ_1 the total deficiency of the one-block components. This settles most classes: at $(10, 2, 0)$, $e = 2$, it gives $5\Lambda \leq 8$ hence $U \leq 1$; at $(15, 1, 0)$ and at $(20, 0, 0)$ with $e \geq 6$ it gives $5\Lambda \leq 20 - e \leq 14$ hence $U \leq 2$; and at $(20, 0, 0)$ with $e \in \{1, 2\}$, $c_1 \geq 0$ in (2) gives $U = a_2 + 2a_3 \leq e \leq 2$. Only $(20, 0, 0)$ with $e \in \{3, 4, 5\}$ remains, where (5) gives merely $\Lambda \leq 3$. Suppose $U = 3$ there. Then $c_2 = 0$ and $(a_2, a_3) \in \{(3, 0), (1, 1)\}$, and $\Delta_1 \leq D - e - 5\Lambda = 5 - e$; every short one-block component contributes at least 2 to Δ_1 , so $n_S \leq \lfloor (5 - e)/2 \rfloor = 1, 0, 0$ at $e = 3, 4, 5$. But $P = 28$ gives $a_1 = 28 - e - a_2 - a_3$ and $n_{45} = a_1 - n_S$, and $X4h = 0$ gives $\lambda \leq a_2 + a_3 + 1$, so the run cap (4) rearranged as $a_1 - 4\lambda \leq 5n_S$ demands $n_S \geq 2, 1, 1$ when $(a_2, a_3) = (3, 0)$ (there $\lambda \leq 4$, $a_1 = 25 - e$) and $n_S \geq 3, 2, 2$ when $(a_2, a_3) = (1, 1)$ (there $\lambda \leq 3$, $a_1 = 26 - e$). All six cases contradict the ceiling on n_S , so $U \leq 2$ at $Z = 0$.

Now let Σ lie in the defect class $(15, 0, 1)$, $e = 2$, where $X4h = 0$ and Lemma 3.13 leaves two skeletons, each with three hop-chains, no pointer cycle, one two-block component and one further non-critical run at the defect transition. In the zh skeleton $[nc_H K_1] + [K_2 nc_2 K_3]$ the run nc_2 is entered by the non-link ending the chain of K_2 , and the hit leaving nc_2 enters the chain head K_3 : one span arc. The run nc_H is entered by the defect heavy transition, which leaves a completion and so ends its chain, and the hit leaving nc_H enters the chain head K_1 : the defect arc. In the $zp3$ skeleton $[K_1 nc_1 K_2] + [K_3 nc_2]$ the run nc_1 gives one span arc in the same way, and nc_2 is entered by the non-link from the last path of K_3 , which ends that chain, while the premature cost-3 exit of the defect lands on a critical permutation that is not hop-entered and hence heads a chain: the defect arc. In both skeletons the remaining chain head is the path of q_1 and the remaining chain end is the global end, so no other transition ends one chain and enters another, and CG has three vertices and two arcs. In- and out-degrees are at most one, the chain of q_1 has no in-arc and the chain of the global end has no out-arc, so CG is one directed path plus vertex-disjoint cycles, every cycle contains the span arc or the defect arc, and there are at most two cycles. Hence $U = 2$ in the defect class, within the engine's two-loop budget. $\square$

The engine searches at most two loops because of this lemma, not because the census emits no configuration with more.

Theorem 4.8 (Coverage). *For every superpermutation Σ of length 871 there are a class κ among the sixteen, a configuration γ emitted by the census for κ (or a wiring of Lemma 3.13), an instance ι among the 631, and*

a mode $m \in \{0, 1\}$ such that (ι, m) is one of the 1111 committed runs and, after relabelling, the placement Σ induces on the 144 ρ -orbits is accepted by that run.

Proof. Theorem 3.10 gives κ together with waste = 146 and slack = 0, and zero slack makes the re-entry count vanish, so Lemma 4.1 applies at $Z = 0$ and Lemma 3.13 at $Z = 1$. Every rejection test of the census is necessary for real orderings by Lemma 4.6, and the expansion applies none, so γ and ι are emitted. Lemma 4.1 at $Z = 0$, and the two skeletons of Lemma 3.13 in the defect class, make Σ induce exactly the geometric data the model prescribes; every cycle is type I by Lemma 4.3, so the endgame’s placement is exhaustive, and every cost- ≥ 4 join has cost exactly 4 by Lemma 3.12, so it lies in the engine’s move set. By Lemma 4.7 the closure graph of Σ is one path plus at most two loops: with no loop, mode 0 searches its structure; otherwise mode 1 anchors a loop-carrying piece at 123456, legitimate because relabelling acts transitively, searches the loop content, optionally anchors a second loop, and runs the linear remainder. Mode 1 is run for exactly the 480 instances carrying a loop-capable piece; the other 151 have $U = 0$. $\square$

Proposition 4.9. *All sixteen classes of Theorem 3.10 are empty.*

Proof. The searcher is Section 5.8. All 1111 runs, 631 in linear mode and 480 in loop mode, terminate with UNSAT EXHAUSTED and no solution, visiting 209,483,950,989 nodes in total, with no run stopped by a resource cap. By Theorem 4.8 a length-871 superpermutation would be accepted by one of them, so none exists. $\square$

Proof of the Main Theorem. Suppose a superpermutation of length 871 existed. By Theorem 3.10 its structural class is one of sixteen. Proposition 4.9 eliminates all sixteen classes, and Proposition 4.5 independently eliminates the six largest. So none has length exactly 871, and by Lemma 2.6 none has length below 871 either. Hence $M(6) \geq 872$, and with Proposition 2.2, $M(6) = 872$. $\square$

5 The programs

A search that reports no solution proves an emptiness statement only if the enumeration it exhausts contains every real candidate, which is the mathematics above, and if the program that ran is the program the mathematics describes, which this section documents program by program. Every program is a file in [11], needs only a C compiler and Python 3, and prints its expected output at the top of its own `run.sh`; `VERIFY.md` lists the command and expected output for each.

Table 2: The assisting programs. Times are wall clock on one Apple M4 with ten cores, measured in a complete re-execution on 19 September 2026 in which every stage reproduced the committed results, the searches node-identically; run back to back the stages took 5 h 51 m.

§	program	source file	time
5.1	membership checker	<code>src/checker/verify6.c</code> (C) <code>src/checker/checker.py</code> (Py)	< 1 s
5.2	capacity enumerator	<code>src/frontier/chains871.noprune.c</code> (C) <code>src/cleanroom-frontier/sweep.decide.c</code> (C)	1 h 11 m 2 min
5.3	class scanner	<code>src/class-scanner/scan871.py</code> (Py) <code>src/class-scanner/elimtable.py</code> (Py)	44 s
	control batteries	<code>src/audit/checks/itemB.controls.py</code> (Py) <code>src/external-872-control/run.sh</code>	< 1 min
5.4	ordering search	<code>src/ordering-search/e9exact.c</code> (C)	2 h 18 m
5.5	finite-lemma checkers	<code>src/finlem/checks/lem58.k1.py</code> (Py) <code>src/finlem/checks/lem59.z1.py</code> (Py) <code>src/finlem/checks/lem67.c2.py</code> (Py)	< 1 min
5.6	chain query, covering	<code>src/chain-query/chainq.c</code> (C) <code>src/covering-test/cover.c</code> (C)	12 min
5.7	census generator	<code>src/coverage/checks/census_gen.py</code> (Py) <code>src/coverage/checks/loopbound.py</code> (Py)	5 s
5.8	elimination engine	<code>src/class-elimination/mysweep.c</code> (C) <code>src/engine-original/realize.c</code> (C, linear mode) <code>src/cleanroom-loop/src/n6c11.decide.c</code> (C) <code>src/testbattery/battery.py</code> (Py)	1 h 01 m 2 min 51 min 4 min

5.1 The membership checker

Purpose. Proposition 2.2, on `data/best872.txt`. *Output:* the line `length=872 distinct_perms=720 VALID_SUPERPERM6` from each implementation. *Method:* slide a width-6 window along the string and count the distinct windows that are permutations. *Trust:* two independently written implementations agree.

5.2 The capacity enumerator

Purpose. Lemma 3.3. *Output:* the tables g and g_S , the typed maxima of $m - 2c$ behind Lemma 3.3(4), and an attaining chain per non-empty cell. *Method:* depth-first extension of abstract chains from the head 123456, one segment at a time, without dominance pruning. *Trust:* two independently written enumerators agree entry for entry, one of them pruning-free at 9.94×10^{10} nodes, and `src/frontier/chains871_witness.c` exhibits an attaining chain for all 80 non-empty cells, so no entry consumed is an unattained bound.

5.3 The structural-class scanner

Purpose. Theorem 3.10. *Output:* the surviving classes and a committed elimination table giving for each of the thirty-four types the deciding test and its two sides. *Method:* enumerate every profile of the thirty-four types and reject on any of (S1)–(S14). *Trust:* a second independently written implementation agrees on the survivor set, the printed table makes the survivors derivable by hand, and the battery `src/audit/checks/itemB_controls.py` evaluates all fourteen tests on 97 real superpermutations of length at most 873 with zero rejections, many of them tight. As an external control, the structurally different length-872 witness published in [9] was run through the same batteries by `src/external-872-control/run.sh` and is rejected by no rule; the transcript is committed, and the string, being third-party material, must be supplied by the reader.

5.4 The ordering search

Purpose. Lemma 3.11. *Output:* per instance, a line `solutions=0 ... EXHAUSTED` with its node count. *Method:* depth-first extension of an ordering with the given segment structure, with neighbour tables for every cost from 3 to 6; its prunes are arithmetic and Lemma 3.3(1). *Result:* nine instances, 8.95×10^{11} nodes, none accepting, none capped. *Trust:* a second implementation agrees on all nine verdicts and node counts.

5.5 The finite-lemma checkers

Purpose. Lemmas 3.12, 3.13 and 4.3. *Output:* the surviving configurations, with one rejection witness per rejected candidate; the committed results are 0 of 2402, 2 of 24 and 0 of 241. *Method:* walk the component censuses the class admits and evaluate the deficiency floors, the pinch (3)–(4) and the packing test of Lemma 3.5. *Trust:* a discrimination control: re-run at $X4h = 2$ instead of $X4h = 1$, the first checker leaves four survivors rather than none.

5.6 The chain query and the covering test

Purpose. Proposition 4.5. *Output:* the exact maximum member count of a chain meeting a prescription with pinned endpoint lengths, and the number of partitions of the leftover classes into ρ -segments of length four. *Method:* the enumeration of Section 5.2 with the endpoint lengths pinned, and exact cover by depth-first search. *Result:* 5544 queries in the elimination battery, and `covers=0` on all 59 configurations of the four decisive runs. *Trust:* the same sweep on the shape of the real length-872 object returns `covers=1`; the query agrees with the certified tables on all 186 prescriptions of cost at most 16; second implementations of both agree.

5.7 The census generator

Purpose. Theorem 4.8, through Lemma 4.6. *Input:* the fifteen $Z = 0$ classes and, for the expansion only, the two defect wirings of Lemma 3.13. *Output:* 200 configurations from the $Z = 0$ classes, 631 instances, 441 from them and 190 from the defect wirings, and 1111 runs, printed in full. *Method:* enumerate the configuration

tuple over the ranges forced by (2), apply the tests (C1)–(C11), emit the rest; expand each survivor into concrete piece multisets with no test applied at all. *Trust*: a second implementation, written from the specification and enumerating a different space, is set-identical, and `src/coverage/checks/match_runs.py` checks the instance list against the committed run log by exact set equality.

5.8 The class-elimination engine

Purpose. Proposition 4.9, the decisive computation of the paper. *Output*: per instance and mode, an accepting placement or the marker `UNSAT EXHAUSTED` with node and closure counts, committed in `CERTIFICATES/sweep_all.log`. *Method*: place the pieces on the 144 ρ -orbits one at a time, branching over orbit positions, the six hop targets at each chain junction, the 24 cost-4 successors at each heavy join, and, in loop mode, the anchored closure structures of Lemma 4.7; an exact-cover endgame places the type-I cycles. *Result*: 1111 runs, 2.09×10^{11} nodes, every one exhausted, none accepting, none capped.

Trust. *Dual implementation*: the 631 linear runs are carried by a second engine, `src/engine-original`, node- and closure-identically, and the 480 loop runs by a third, `src/cleanroom-loop`, written from the specification alone, which agrees on all 480 verdicts. *Acceptance controls*: the engine accepts the real length-872 object’s configuration in linear mode, a constructed loop-carrying placement in loop mode, and, in a variant extended with type-II placement, a planted type-II configuration. *Discrimination* (`src/testbattery`): replacing the proved six-target hop rule by “a hop may land anywhere” flips one instance from exhausted to accepting in linear mode and to non-terminating in loop mode. *Reproducibility*: every one of the 1111 runs reproduces from an independent build with exact node and closure counts, in `CERTIFICATES/reexecution/`.

The first engine differs from the specification in three respects, each accounted for and none affecting a verdict. First, it inlines the next hop from the exit of the defect piece rather than returning through the linear driver, so when that piece is placed last the endgame branch at its exit is not among the states it visits; the specification-following engine does visit that branch and returns all 190 defect-class instances in linear mode exhausted. Second, its heavy pass omits the defect-piece move. Third, its cost-4 closure is available only at span anchors. The last two are vacuous, since the defect piece occurs only in the class (15, 0, 1), where $X4 = 0$ and neither the heavy pass nor a cost-4 closure is ever reached.

5.9 What the programs together establish

A run that exhausts without accepting proves a class empty exactly when every candidate lies in the engine’s initial set, every valid extension is generated and no valid branch is pruned, and every complete candidate is accepted: Theorem 4.8, the model of Section 4, and the acceptance controls. `VERIFY.md` carries these as twelve numbered obligations.

One limitation should be stated plainly. The exclusion of type-II cycles and three-block components has no positive control from a real object, since no known superpermutation exhibits either; it rests on the proof of Lemma 4.3. What the reader still trusts, beyond the mathematics, is that the programs that ran are the programs specified, on the evidence of Section 5.8; we state this rather than call the result unconditional.

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